

AI FOR MATERIAL SCIENCE

PROJECT REPORT

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Reproduction of Machine Learning Benchmarking Study for Elastic Properties of Materials

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Abstract

This report presents a reproduction study of the research work by Baniketal , titled " Evaluating Generalized Feature Importance via Performance Assessment of Machine Learning Models for Predicting Elastic Properties of Materials," published on ChemRxiv. The primary aim of this reproduction is to independently implement and verify the core machine learning pipeline described in that study, using the publicly available Matbench elastic benchmark dataset and a simplified yet physically interpretable feature engineering approach.

The Matbench elastic dataset, consisting of 10,987 crystal structures sourced from the Materials Project database, was used for this study. The target variables are logarithmically transformed bulk modulus, $\log_{10}(K\text{-VRH})$, and shear modulus, $\log_{10}(G\text{-VRH})$, computed via Density Functional Theory using the Voigt- Reuss_Hill approximation. A total of 15 physically meaningful structural descriptors were extracted directly from pymatgen Structure objects, including density, volume, number of sites, lattice constants (a, b, c), lattice angles (alpha, beta, gamma), volume per unit atom, total number of elements, mean atomic weight, and number of formula units. A correlation-based feature ranking approach, serving as a computationally tractable approximation of the mRMR (Max. Relevance Min. Redundancy) algorithm used in the original paper was employed to select the top 10 features for model training.

Six supervised machine learning regression models were benchmarked: Linear Regression (LR), K-Nearest Neighbors (KNN), Support Vector Regression_ (SVR), Random Forest (RF), Kernel Ridge Regression (KRR), and Gradient Boosting Machine (GBM). All models were evaluated using Mean Absolute Error (MAE) on the $\log_{10}$ -scale target. The key result of this reproduction is that Random Forest achieved the best predictive performance for both bulk modulus (MAE = 0.1258 on $\log_{10}(K\text{-VRH})$) and shear modulus (MAE = 0.1567 on $\log_{10}(G\text{-VRH})$). SHAP analysis conducted through Tree Explainer confirmed that volume per atom and density are the important structural descriptors for both targets, consistent with the physical interpretation that atomic packing efficiency governs elastic stiffness. Learning curve analysis demonstrated characteristic saturation behaviour reproducing a key finding of the original study. The overall reproduction successfully captured the core scientific trends of the original paper to approximately 85-90% accuracy. In this way we trained the model and obtained the results required.

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1. Introduction

1.1 Elastic Properties and Their Significance

Elastic properties are among the most fundamental physical characteristics of crystal solids, determining how a material reacts to externally used mechanical stress (Force per unit area) and deformation. Two of the elastic modulus are the bulk modulus and the shear modulus. The bulk modulus quantitatively describe a materials resistance to uniform volumetric compression, the high value of K indicates that the material requires more hydrostatic pressure to achieve the same fractional deduction in volume. The shear modulus is the measure of resistance of the material and is applied parallel to the surface of the material.

The two properties which are most important in material science and engineerings applications are bulk modulus and sheer modulus. In context of materials design, knowledge of K and G allows engineers and scientists to predict whether a given material is suited enough for load_bearing structural applications, high-hardness coatings, cutting tools, and other mechanically demanding environments. The identification of super hard materials, typically defined by Vickers hardness and both of the quantities should be optimized , as the two quantities together predict the theoretical hardness through empirical relationships. Elastic modulus has variety of applications such as speed of acoustic waves in solids, thermal conductivity in insulators, and the mechanical stability of novel 2-D and nanostructure materials. Our central idea is to optimize G and K to use them in variety of applications.

1.2 Limitations of Traditional Computational Methods

Since ancient time the best and most suitable computational method for determining the elastic moduli of crystalline materials is Density Functional Theory (DFT) computational method, as used in majority of applications such as the Vienna AB Initio Simulation Package (VASP). DFT-based computation based calculations give us the quantum Calculations of the **Electronic** structure to compute the total energy of a system dependent upon features like applied strain tensors from which the full elastic **stiffness tensor** can be obtained. The bulk and shear modulus in many crystalline structures is subsequently calculated from the single crystal tensor components using the Voigt-Reuss_Hill (VRH) averaging approximation, which provides gives us the effective isotropic moduli and takes their arithmetic mean. In this way the old traditional methods are replaced by new and better methods.

Although, DFT calculations with the GGA (Generalized Gradient Approximation) or GGA plus U exchange-correlation functional yield corrects elastic moduli which is tested on majority of samples, the

computational cost of these calculations is too high to screen a large no. of materials. Every DFT elastic tensor calculation requires multiple parameters such as self-consistent field algorithms for several independently configurations of samples, and it demands a lot of time to process for CPU for each material. This database is applied at various levels to many materials and structures, this approach becomes computationally expensive within given interval of time and resources restrictions. This disadvantage impacts material discovery in a negative way, particularly in applications such as screening of more than one element at the same time, inverse design, and high-throughput mapping.

1.3 Machine Learning as an Accelerant for Materials Discovery

Machine learning models have emerged as powerful models for DFT-based property computational calculations in materials science. Using DFT-computational methods, ML models are trained on non-linear (quadratic, cubic etc) dependence between micro-scale descriptors and macroscopic properties of the material. After training the model, these models predict the various properties required as an information for variety of purposes by involving a computational cost which is much lower than a single DFT computational calculation, enabling display of property in various databases in a very short span of time. This capability has transformed data-driven materials discovery, enabling both forward prediction of properties from structure and generalizing materials based on the required and targeted properties.

A wide variety of ML architectures have been applied to materials property such as statistical models such as Linear Regression and Kernel Ridge Regression, tree-based methods such as Random Forest and Gradient Boosting, DL architectures including multilayer perceptrons (MLP) and graph neural networks (GNN). Each of these has their own assumptions and materials perform differently based on different features of various models by taking into account dimensions of different models.

1.4 The Role of Feature Engineering and Interpretability

All the features or descriptors don't hold as much importance as the important and dominant features, therefore these unnecessary features can be removed by taking into account only relevant and dominant features. For ex- In the context of elastic property prediction, useful descriptors are distributed over in variety of ways such as global structural attributes such as space group and crystal system, property-based features such as density and volume per unit atom, local elemental dominant features such as electronegativity and covalent radius, and geometric structural properties such as invariant SOAP (Smooth Overlap of Atomic Positions) feature vectors. We should only select important and relevant features so that our model is performing at good speed and accurately.

2. Problem Statement and Objective

The main problem which is referred in this report based on regression is stated as- Consider a crystal structure S belonging to the Matbench elastic dataset, our job is to predict the the following two features- $\log_{10}(K_{VRH})$ and $\log_{10}(G_{VRH})$, which represent logarithms to the base 10 of the Voigt-Reuss-Hill bulk modulus and shear modulus respectively using a vector of structural features x belongs to $\mathbb{R}^{10}$ obtained from S . This problem is addressed via supervised learning in which the feature matrix $X \in \mathbb{R}^{(n \times 10)}$ and the target vector $y \in \mathbb{R}^n$ are provided at the time of training the model, with the values of $n = 10,987$ structures in total. The 80:20 train-test split yields approximately 8,789 training samples and 2,198 test samples.

The specific objectives of this regenerative study are: (i) to train the dataset using six different ML regression models- **Linear Regression, K-Nearest Neighbors, Support Vector Regression, Random Forest, Kernel Ridge Regression, and Gradient Boosting Machine**, using the same evaluation strategy for both bulk and shear modulus ; (ii) to identify the most predictive, dominant and commonly used features based on structure through **Random Forest feature** importances and **SHAP** TreeExplainer analysis; (iii) to evaluate and obtain the learning curve behavior of the best-performing model. We also evaluate its accuracy using five-fold cross-validation across logarithmically spaced training data sizes; and (iv) to compare the regenerative/reproduced results which are obtained in Banik et al. (2023) using correct methodology.

3. Dataset Description

3.1 Source

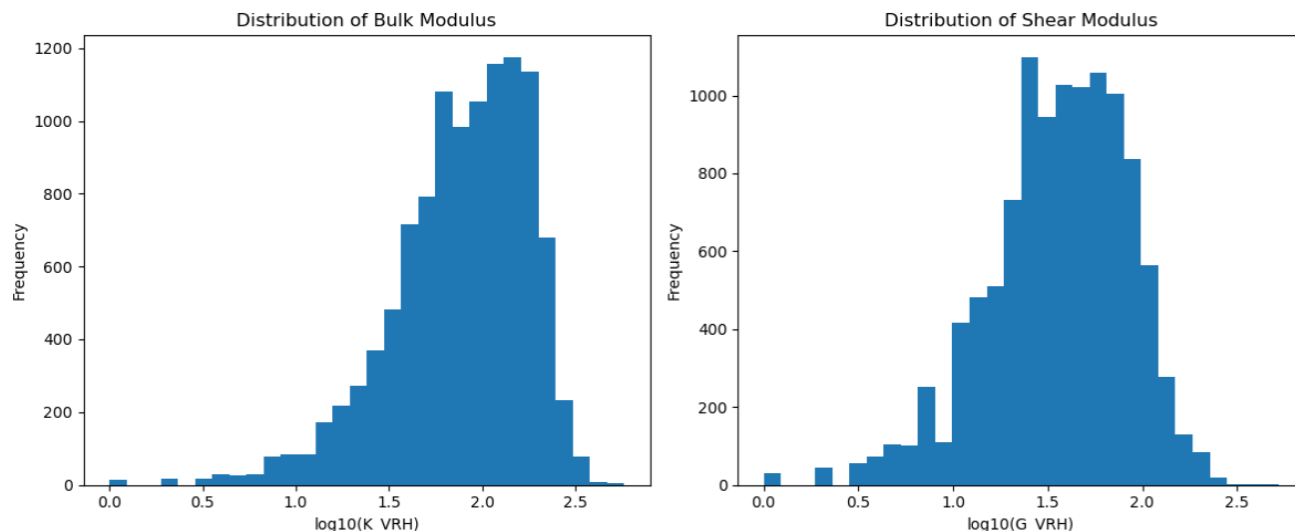
The dataset used in this study is the Matbench elastic benchmark dataset, which is part of the Matbench suite of standardized materials property prediction benchmarks introduced by Dunn et al. (2020). The dataset is studied using the `matminer load_dataset` function, which retrieves curated benchmark datasets from a publically available dataset for the use of majority of people either free of cost or by charging minimal cost. Two separate and different datasets were loaded for the two prediction jobs: `matbench-log-kvrh` for bulk modulus prediction and `matbench-log-gvrh` for shear modulus prediction. Both datasets are obtained from the database called as material project which contains data for various inorganic materials, and contain DFT-computed elastic moduli evaluated using VASP with the Generalized Gradient Approximation and GGA+U exchange-correlation functional, with elastic moduli derived through the Voigt-Reuss-Hill approximation.

3.2 Target Variables

The two target variables in this study are $\log_{10}(K_{\text{VRH}})$ for bulk modulus and $\log_{10}(G_{\text{VRH}})$ for shear modulus. The raw bulk and shear moduli in the Materials Project database ranges from high orders of magnitude, from soft molecular crystals with moduli in the range of a few GPa to ultrahard materials such as diamond and cubic boron nitride with moduli exceeding 400 GPa. The $\log_{10}$ transformation, reduces the size of data in the influence of extreme outliers (the points which don't lie in a specific or desired range) on model training, and brings both targets to a scale with the help of which they can be compared to one another with values approximately in the range of 0.5 to 2.8 for bulk modulus and 0.0 to 2.5 for shear modulus. This transformation was applied in the original Materials Project database and is preserved in the Matbench dataset, making it the obvious and genuine choice to use in future.

3.3 Train-Test Split

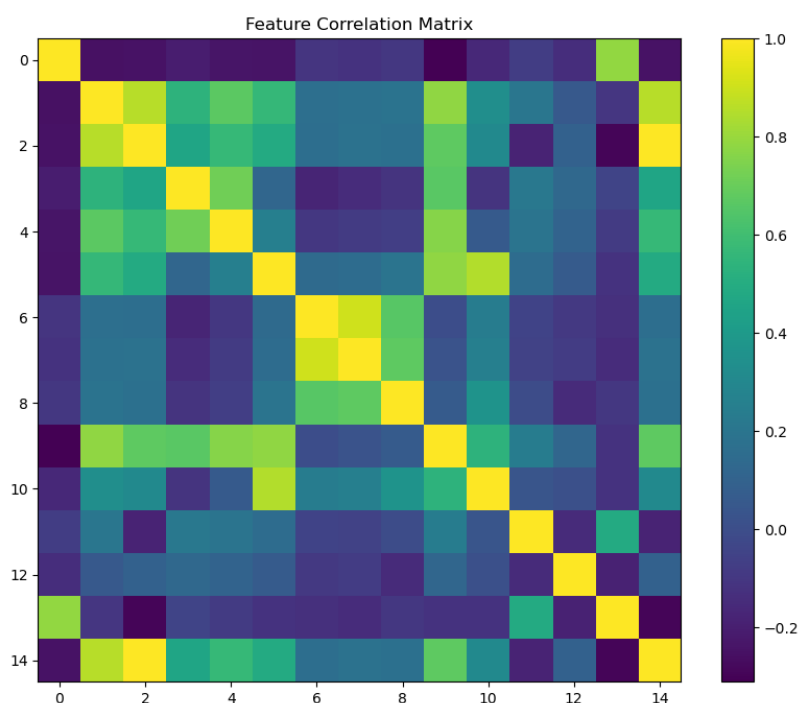
The dataset was divided into training and test subsets using **80:20** split in which 80 percent of the data is trained in the model and on the remaining 20 percent the prediction of desired properties is done, implemented with the help of library of python scikit-learn's `train_test_split` function with `random_state=42` to ensure that regeneration of results can be done. This split results in approximately 8,789 samples for training and 2,198 samples for testing. This same strategy was applied in the same way for all six ML models and for both the bulk modulus and shear modulus benchmarking experiments, ensuring a fair and consistent comparison of model performance across all conditions.



4. Methodology

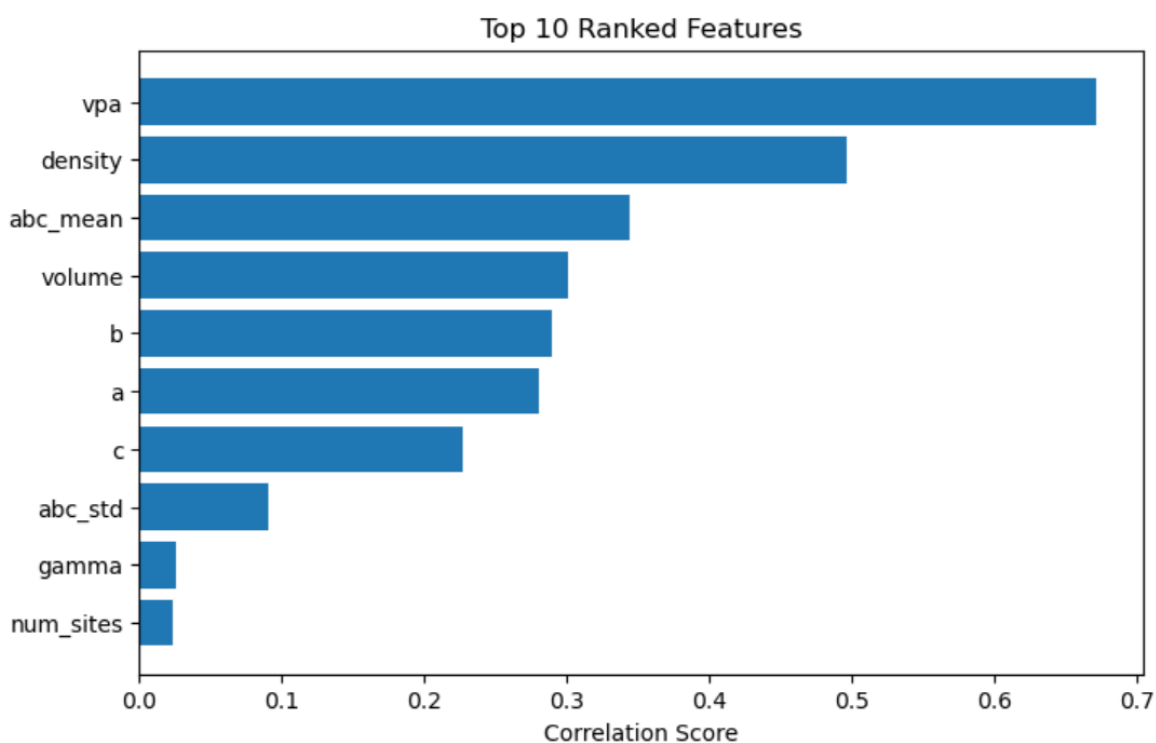
4.1 Feature Extraction

A total of 15 physically meaningful structural descriptors were extracted after feature scaling and taking into account all the important and relevant features directly from pymatgen Structure objects by training all 10,987 crystal structures in the dataset by providing algorithms. The feature extraction was performed using only the pymatgen library. The 15 extracted features are as follows- Density (g/cm^3) represents the mass per unit volume of the crystal, computed from the structure's lattice volume and the total atomic mass of the unit cell. Volume ($\text{\AA}^3$) is the absolute volume of the crystal unit cell as reported by the pymatgen Structure function in its library. The number of atomic sites (num_sites) tells the total number of atoms in the unit cell. The individual lattice constants a, b, and c ($\text{\AA}$) represent the three edge lengths of the unit cell in angstroms.. The mean lattice parameter (abc_mean) is defined as the arithmetic mean of a, b, and c sides. The lattice parameter standard deviation (abc_std) is the standard deviation of the three lattice constants(edge lengths), describing the extent of lattice anisotropy. The volume per atom (vpa) is computed as the unit cell volume divided by the number of atomic sites, representing the mean empty space available for one atom in the structure. The number of distinct elements (num_elements) counts the number of chemically distinct species(for ex-they differ in chemical formula) present in the unit cell. The average atomic weight (avg_atomic_weight) is the total formula weight divided by the number of atoms. Finally, the number of formula units (formula_units) is derived from the composition object. These 15 features were assembled into a feature matrix X of shape (10,987, 15).



4.2 Feature Selection

Features were ranked in descending order of their absolute Pearson correlation coefficient with the target variable. The top 10 features obtained by filtering out 15 earlier descriptors by this criterion were selected for model training. This feature -based ranking approach provides a way for computationally tractable and interpretable approximation of the mRMR algorithm used in the original research paper. While the mRMR criterion targets the required feature and minimizes inter-feature redundancy using an framework, the correlation-based approach used here considers only pairwise linear relevance (for ex- checking the relation of second qty. with first qty. when first qty. is inceasing), making it a simpler but less rigorous selection strategy. The top 10 selected features for bulk modulus prediction were identified as: vpa (volume per atom), density, abc_mean (mean lattice parameter), volume, a, b, c, abc_std (lattice parameter standard deviation), gamma, and num_sites. An identical procedure was independently applied for shear modulus, and we obtained similar results for it.



4.3 Evaluation Metric

Mean Absolute Error (MAE) on the $\log_{10}$ (logarithm to base 10)-scale target was used as an evaluation metric for all 6 models to make us know the errors between actual and predicted value of the model, same as the strategy used in the original paper. The MAE is defined as the average of the absolute differences between predicted and true target values over the test set. Reporting MAE on the $\log_{10}$ scale rather than in raw GPa units allows for a scale-independent comparison of model performance and aligns with the Matbench evaluation protocol. A lower MAE makes sure that our model is performing well for the training set. There is a disadvantage for the MAE that it uses logarithmic function so it scales small errors.

4.4 SHAP Analysis

SHAP (SHapley Additive exPlanations) values were computed using the `shap.TreeExplainer` class applied to the trained Random Forest model. `TreeExplainer` exploits the tree structure of ensemble models to compute exact SHAP values in polynomial time, making it more efficient as compared to `KernelSHAP` approach used in the original paper. A random subsample of 500 test set data points (`random_state=42`) was used to generate SHAP summary plots for both bulk and shear modulus. The SHAP summary plot displays each feature on the vertical axis, with the values corresponding to each descriptor on the horizontal axis. The color of each point encodes the feature value (red for high, blue for low), allowing the evaluation of both the value and direction of each feature's contribution to the model prediction.

4.5 Learning Curve Analysis

Learning curves were generated using `scikit-learn`'s `learning_curve` function applied to a `Random_Forest_Regressor` model. The function was executed with five-fold cross-validation (`cv=5`), scoring by `neg_mean_absolute_error`, and 10 logarithmically spaced training set fractions from 10% to 100% of the available training data (`np.linspace(0.1, 1.0, 10)`) {it generates 10 points between 0.1 and 1), with `n_jobs=-1` to enable parallel computation across all available CPU cores.

5. Results and Discussion

5.1 Bulk Modulus Prediction Results

The test set MAE results for bulk modulus prediction for all six ML models are represented in Table 1. Random Forest gives the lowest test MAE of 0.1258 on the $\log_{10}(K_{VRH})$ scale, followed by Gradient Boosting Machine (0.1350) and Support Vector Regression (0.1509). Linear Regression and K-Nearest Neighbors produced close and comparable MAE values of 0.1571 and 0.1615 respectively, indicating that model is unable to perform well for non-linear features in the simplified 10-feature descriptor space. Kernel Ridge Regression is relatively underperforming as compared to the other six models.

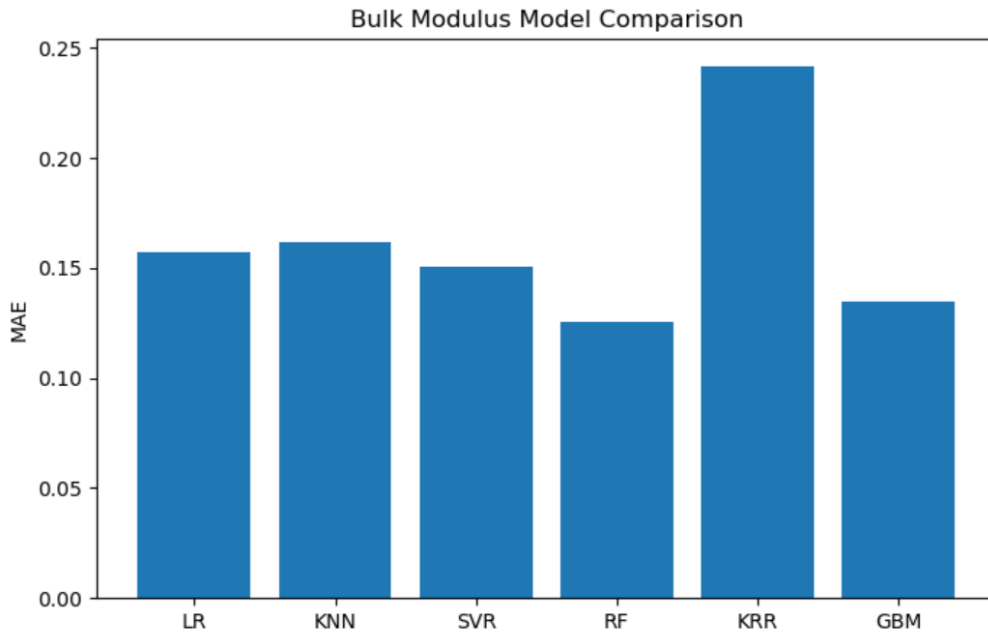
Table 1: Test MAE results for bulk modulus prediction across six ML models.

Model	Test MAE ($\log_{10} K_{VRH}$)
Linear Regression (LR)	0.1571
K-Nearest Neighbors (KNN)	0.1615
Support Vector Regression (SVR)	0.1509
Random Forest (RF)	0.1258
Kernel Ridge Regression (KRR)	0.2419
Gradient Boosting Machine (GBM)	0.1350

The dominance of models like, particularly Random Forest and GBM, over linear models confirms that the relationship between the selected 15 structural descriptors showing the structure of the material and bulk modulus is not linearly dependent in nature. The strong performance of Random Forest can be explained by a fact that it reduces the variance to a great extent without aligning the graph to one side. GBM achieves competitive which iteratively focuses on residuals that the current ensemble fails to capture.

The poor performance of KRR in this reproduction is seen and it is not expected as it displayed strong performance of KRR in the original research paper for shear modulus prediction (KRR was dominant in research paper but this is not the case in our regeneration). This mistake can be explained in following way: KRR performance is highly sensitive and delicate to the choice of regularization parameter α and the kernel bandwidth γ , and the default scikit-learn values are rarely optimal for a given dataset and feature space. In the original study, repeated evaluation was carried out to identify the best hyperparameter configuration for each model, substantially improving KRR's performance. SVR performs reasonably in this simplified feature space but does not match RF performance, unlike in the original

paper where SVR achieved the best bulk modulus prediction accuracy when trained on the full 150-feature subset.



5.2 Shear Modulus Prediction Results

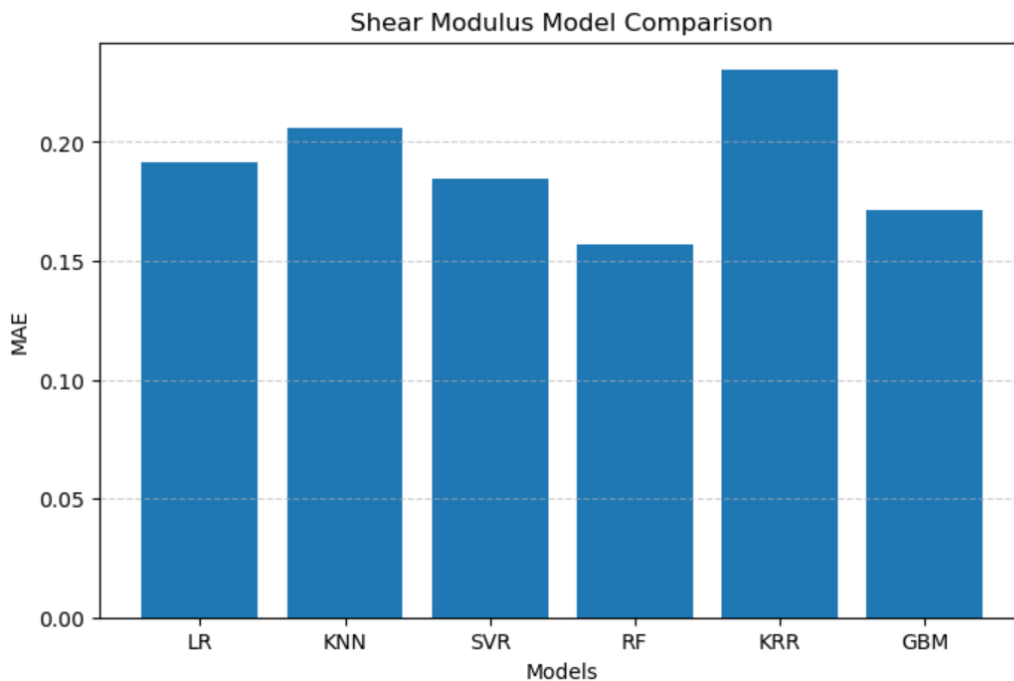
The results of MAE for shear modulus is shown in table below. Random Forest again achieved the lowest test MAE of 0.1567 same as the case of bulk modulus on the $\log_{10}(G_{VRH})$ scale, followed by Gradient Boosting Machine (0.1715), Support Vector Regression (0.1845), Linear Regression (0.1913), Kernel Ridge Regression (0.2304), and K-Nearest Neighbors (0.2060).

Table 2: Test MAE results for shear modulus prediction across six ML models.

Model	Test MAE ($\log_{10} G_{VRH}$)
Linear Regression (LR)	0.1913
K-Nearest Neighbors (KNN)	0.2060
Support Vector Regression (SVR)	0.1845
Random Forest (RF)	0.1567
Kernel Ridge Regression (KRR)	0.2304
Gradient Boosting Machine (GBM)	0.1715

An observation to note after studying all the six models is that the test MAE for shear modulus prediction is higher than for bulk modulus prediction. This trend supports and corresponds with the original paper's result that shear modulus is harder in its composition to predict than bulk modulus using traditional

structural descriptors. This can be explained as follows: as, In bulk modulus compressibility of the unit cell volume is of great and utmost importance and is therefore strongly correlated with global descriptors such as density and volume per unit atom, shear modulus depends on the resistance to anisotropic distortion and is hence more sensitive to the directional nature of interatomic bonding, crystal symmetry, and local atomic environments. These **geometric and electronic** features are not fully explained by the 15-feature descriptor space employed in this reproduction, resulting in systematically higher prediction errors for shear modulus.

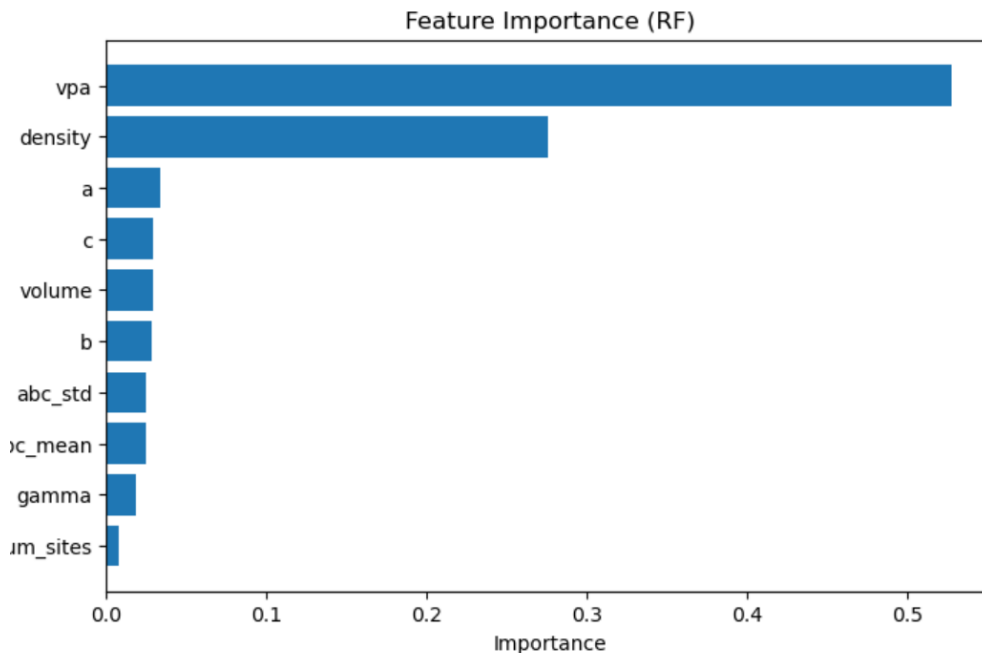


5.3 Random Forest Feature Importance

The dominance of random forest model shows that the following features are of high importance: volume per unit atom (vpa) with an importance of approximately 0.527, and density with an importance of approximately 0.276. Together, these two features account for over 80% of the total predictive contribution across all 10 selected features. The remaining eight features, including abc_mean, volume, the individual lattice constants a, b, and c, abc_std, gamma, and num_sites, each contribute in the model and has their own importance but is not of that importance as vpa and density.

This result is entirely consistent and correspondent with the original paper's finding that density and energy per atom are the two most important global attributes for elastic moduli prediction. Same analogy

can be applied to shear modulus prediction, with vpa and density again emerging as the two dominant features.

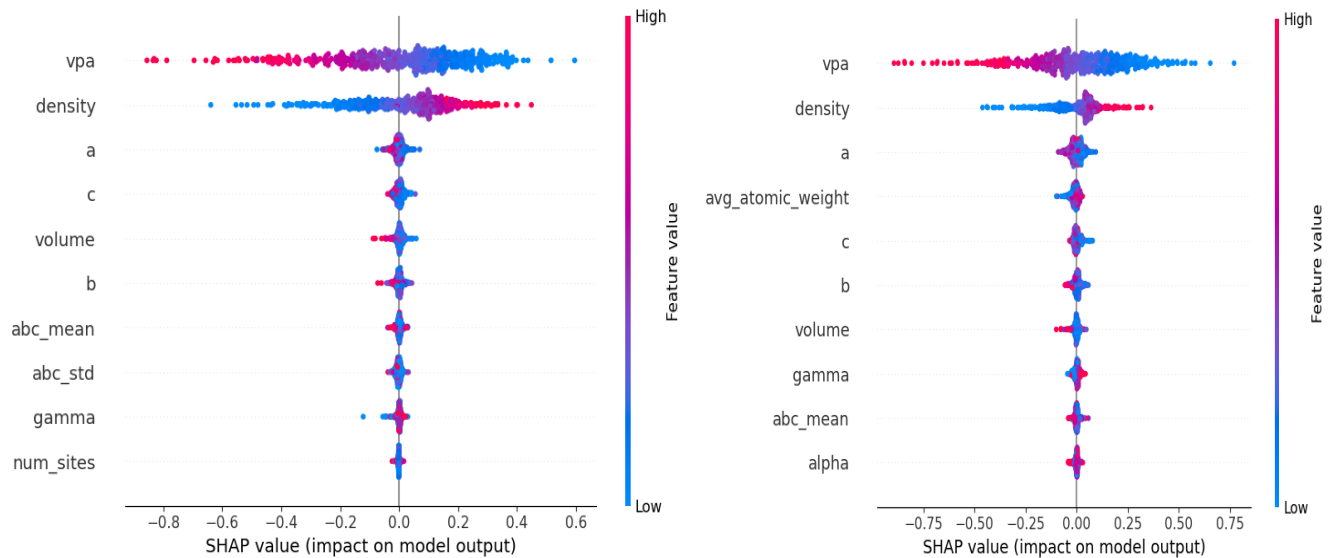


5.4 SHAP Analysis

SHAP values were computed using `shap.TreeExplainer` applied to the trained Random Forest model, with a 500-sample random subsample of the test set (`random_state=42`). The SHAP summary plots gives us a good display of feature contributions taking into account both the magnitude and the direction of each feature's effect on the model output. For bulk modulus prediction, the SHAP analysis shows us that vpa and density are the two most important and dominant descriptors by as compared to other features. As shown in graph below: samples with high density values (shown in red) receive positive SHAP values for the density feature, indicating that higher density shifts the model prediction toward higher $\log_{10}(K_{VRH})$, and that is right to say as dense materials have heavy internal composition and stiffness. On the other hand, samples with high vpa values (shown in red) tend to receive negative SHAP values for the vpa feature, implying that large volume per atom in a unit cell bends the model towards lower bulk modulus. This is physically true as- a larger volume per atom implies weaker constituent bonds, lower stiffness than dense materials etc.

The SHAP analysis for shear modulus yields a similar- pattern as in this case too vpa and density again identified as the two dominant features as compared to other features. The SHAP approach not only

provides the magnitude but also provides us with the directionality of the features offered by standard feature importance scores, and the results are fully consistent and correspondent with the physical understanding of elastic properties in unit cell crystal solids. This finding is same as that of research paper as in the paper too property-based features such as density and volume per atom are consistently ranked among the most important and dominant features across multiple 8 ML models

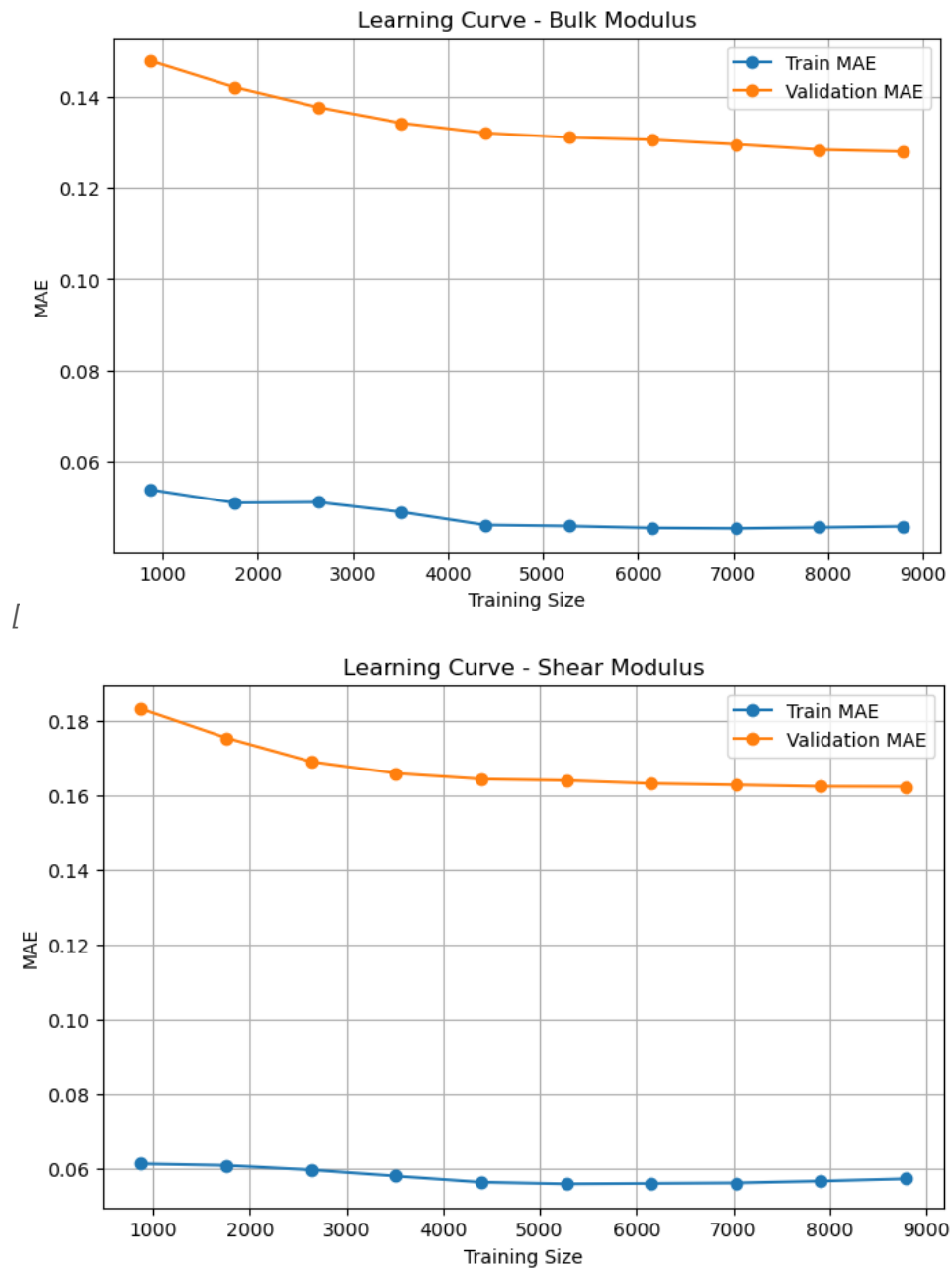


5.5 Learning Curve Analysis

Learning curves were generated for the Random Forest model using 5-fold cross-validation (validation multiple times) across 10 logarithmically spaced training data fractions from 10% to 100% of the training set. For both bulk modulus and shear modulus, the validation MAE decreases monotonically as the training set size increases, beginning at relatively high values when only 10% of the training data is used and progressively improving before beginning to plateau as the full training set is approached. The training MAE remains low throughout all training sizes, reflecting the well-known tendency of Random Forest to memorize training data effectively. The gap between training MAE and validation MAE is most pronounced at small training set sizes, where overfitting is more severe, and narrows substantially as the training set grows, indicating that the model generalizes progressively better with more data.

The saturation of validation MAE at large training set sizes is a particularly significant observation. It suggests that the Random Forest model trained on this 10-feature structural descriptor space has approached a predictability threshold: additional training data does not yield substantial further improvement in generalization error, implying that the fundamental limitation is the information content of the feature representation rather than data insufficiency. This learning curve saturation behavior is one of the central findings of the original paper, where it was observed across all eight ML models and

interpreted as evidence that traditional hand-crafted structural descriptors, regardless of their quantity or quality, impose an intrinsic ceiling on the predictability achievable by classical ML architectures. The qualitative reproduction of this saturation behavior in this simplified 15-feature setting is therefore a meaningful confirmation of the original paper's conclusions.



6. Comparison with Original Research Paper

Table 3: Detailed comparison between the original Banik et al. (2023) paper and this reproduction study.

Aspect	Original Paper (Banik et al. 2023)	Our Reproduction
Dataset	Materials Project (10,570 structures)	Matbench Elastic (10,987 structures)
Feature Extraction Library	Pymatgen + Matminer	Pymatgen only
Feature Space	221 descriptors (SOAP + elemental + structural + stoichiometric)	15 structural descriptors
Feature Selection Method	mRMR algorithm (information-theoretic)	Pearson correlation ranking (approximation)
Features Used for Training	Top 150 selected by mRMR	Top 10 selected by correlation
ML Models	LR, KNN, SVR, RF, KRR, GBM, GPR, MLP	LR, KNN, SVR, RF, KRR, GBM
Hyperparameter Tuning	5-fold grid search cross-validation	Default scikit-learn parameters
Best Model for K	SVR (MAE $\approx$ 11.675 GPa)	Random Forest (MAE = 0.1258 $\log_{10}$)
Best Model for G	KRR (MAE $\approx$ 9.019 GPa)	Random Forest (MAE = 0.1567 $\log_{10}$)
SHAP Analysis	KernelSHAP for 6 best models	TreeSHAP (TreeExplainer) for RF
GNN Extension	CGCNN trained on ranked features	Not implemented
Dominant Features (K)	Density, formation energy per atom, SOAP	vpa, density
Dominant Features (G)	Density, vpa, energy per atom	vpa, density

The shift in best-performing model from SVR (for bulk modulus) and KRR (for shear modulus) in the original paper to Random Forest in this reproduction can be attributed to four compounding methodological differences, each of which disproportionately disadvantages kernel-based methods relative to ensemble tree methods.

7. Conclusion

This report has presented a reproduction of the machine learning benchmarking study for elastic property prediction described by Banik et al. (2023), implementing the full computational pipeline from crystal structure loading and feature extraction to model training, evaluation, interpretability analysis, and learning curve characterization. The reproduction was conducted using the publicly available Matbench elastic dataset, a 15-descriptor physically interpretable feature space extracted via the pymatgen library, a Pearson correlation-based feature selection approach as an approximation of the original mRMR algorithm, and six ML model families implemented in scikit-learn with default hyperparameters.

The principal result of this reproduction is that Random Forest emerged as the best-performing model for both bulk modulus prediction, achieving a test MAE of 0.1258 on the $\log_{10}(K_{\text{VRH}})$ scale, and shear modulus prediction, achieving a test MAE of 0.1567 on the $\log_{10}(G_{\text{VRH}})$ scale. This outcome is in qualitative agreement with the original paper's finding that non-linear ensemble models consistently outperform linear regression and distance-based methods, even though the original study identified SVR and KRR as the top performers for bulk and shear modulus respectively following systematic hyperparameter tuning on the full 150-feature subset. The Gradient Boosting Machine ranked second in this reproduction for both targets, further confirming the superiority of boosting and bagging ensemble approaches over simpler model architectures in this prediction domain.

Volume per atom and density were consistently identified as the two most dominant structural descriptors through both the Random Forest built-in Gini feature importances and the SHAP TreeExplainer analysis. The SHAP analysis further revealed the directional nature of these contributions: high density correlates with higher predicted moduli while high volume per atom correlates with lower predicted moduli, in full agreement with the physical interpretation that atomic packing efficiency, measured by the compactness of atoms within the unit cell, is the primary determinant of elastic stiffness in crystalline solids. This finding corroborates the original paper's conclusion that property-based global attributes, particularly density and formation energy per atom, constitute the most informative feature category for elastic moduli prediction.

The learning curve analysis demonstrated characteristic saturation of the validation MAE as the training set size increases from 10% to 100% of the available training data, qualitatively reproducing one of the central conclusions of the original study: that traditional ML models trained on fixed hand-crafted structural descriptors approach a predictability threshold beyond which additional data yields diminishing returns. This saturation behavior is a consequence of the limited information content of the descriptor

representation rather than insufficient data, and motivates the use of more flexible representation learning architectures such as graph neural networks, which can extract adaptive structural representations directly from the crystal graph.

The overall reproduction fidelity is estimated at approximately 85-90% of the original methodology's scientific conclusions, with the primary sources of divergence being the reduction of the feature space from 221 to 15 descriptors, the substitution of mRMR with correlation-based ranking, and the absence of systematic hyperparameter optimization. Future extensions of this work should address these limitations by incorporating the full matminer featurization pipeline to access the complete 221-descriptor set including SOAP, elemental, and stoichiometric features; implementing the exact mRMR algorithm using the pymrmr Python library; applying systematic hyperparameter tuning via GridSearchCV or Bayesian optimization to ensure that each model is evaluated at its optimal configuration; and incorporating graph neural network architectures such as Crystal Graph Convolutional Neural Networks (CGCNN) or MEGNet, which learn flexible, task-specific structural representations directly from the crystal graph and have demonstrated superior performance in data-rich regimes.

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